

Worksheet — 2.3 Algorithms — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Answer key

Activity 1 — Big O match

Algorithm	Best	Average	Worst
Linear search	$O(1)$	$O(n)$	$O(n)$
Binary search	$O(1)$	$O(\log n)$	$O(\log n)$
Bubble sort (with swap flag)	$O(n)$	$O(n^2)$	$O(n^2)$
Insertion sort	$O(n)$	$O(n^2)$	$O(n^2)$
Merge sort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$
Quick sort	$O(n \log n)$	$O(n \log n)$	$O(n^2)$

Note: bubble sort best case is $O(n)$ **only with a swap flag** on already-sorted data. Quick sort worst case is $O(n^2)$ with a poor pivot (e.g. already-sorted data, first/last element pivot).

Activity 2 — Sort trace

BUBBLE SORT of [5, 1, 4, 2, 8] (compare adjacent pairs, swap if left > right; largest bubbles to the end each pass):

Pass	List state after this pass	Swaps this pass
Start	5 1 4 2 8	—
1	1 4 2 5 8	$5 \leftrightarrow 1$, $5 \leftrightarrow 4$, $5 \leftrightarrow 2$ = 3 swaps
2	1 2 4 5 8	$4 \leftrightarrow 2$ = 1 swap
3	1 2 4 5 8	0 swaps → with swap flag, algorithm stops here

Final sorted list: 1 2 4 5 8

Pass-by-pass detail of pass 1 (each adjacent comparison): - (5,1) → swap → 1 5 4 2 8 - (5,4) → swap → 1 4 5 2 8 - (5,2) → swap → 1 4 2 5 8 - (5,8) → no swap → 1 4 2 5 8

INSERTION SORT alternative (acceptable if student chose insertion) — list after each pass (element inserted into sorted front):

Pass	List state after this pass
Start	5 1 4 2 8
1 (insert 1)	1 5 4 2 8
2 (insert 4)	1 4 5 2 8
3 (insert 2)	1 2 4 5 8
4 (insert 8)	1 2 4 5 8

Final sorted list: 1 2 4 5 8 (same result).

Activity 3 — Binary search for 27

List: indices 0–8, values 3, 7, 11, 15, 19, 23, 27, 31, 35.

Step	low	high	mid (index)	value at mid	< = > target?
1	0	8	4	19	$19 < 27 \rightarrow$ search right, $\text{low} = \text{mid} + 1 = 5$
2	5	8	6	27	$27 = 27 \rightarrow$ FOUND

Found at index: **6**. Number of comparisons: **2**.

(Working: step 1 $\text{mid} = (0+8) \text{ DIV } 2 = 4$; step 2 $\text{mid} = (5+8) \text{ DIV } 2 = 6$.)

Activity 4 — Justifications

(a) Binary search is $O(\log n)$ because **each comparison halves the size of the list still to be searched** — it discards the half that cannot contain the target. The number of times n can be halved before reaching 1 is $\log_2(n)$, so the number of comparisons grows logarithmically with n . (Doubling the list size adds only one extra comparison.)

(b) Bubble sort is $O(n^2)$ because it uses **two nested loops**: an outer loop of up to n passes, and for each pass an inner loop making up to n adjacent comparisons. That gives roughly $n \times n = n^2$ comparisons, so time grows with the square of the input size.

(c) Binary search requires the list to be **sorted**; linear search works on unsorted data.

Activity 5 — Tree traversals

Tree: root 40; left subtree 20 (children 10, 30); right subtree 60 (children 50, 70).

- **Pre-order** (Node, Left, Right): 40, 20, 10, 30, 60, 50, 70
- **In-order** (Left, Node, Right): 10, 20, 30, 40, 50, 60, 70
- **Post-order** (Left, Right, Node): 10, 30, 20, 50, 70, 60, 40

Ascending order is produced by the **in-order** traversal of a BST.

Activity 6 — Dijkstra A to F

Edges (undirected): A-B 4, A-C 2, B-C 1, B-D 5, C-D 8, C-E 10, D-E 2, D-F 6, E-F 3.

Settle the unvisited node with the smallest tentative distance each round; relax its neighbours.

Settled (order)	A	B	C	D	E	F
Initial	0	∞	∞	∞	∞	∞
Settle A (0)	0	4	2	∞	∞	∞
Settle C (2)	0	3	2	10	12	∞
Settle B (3)	0	3	2	8	12	∞
Settle D (8)	0	3	2	8	10	14
Settle E (10)	0	3	2	8	10	13
Settle F (13)	0	3	2	8	10	13

Relaxation notes: - From A: B = 4, C = 2. - Settle C (2): B = $\min(4, 2+1) = 3$; D = $2+8 = 10$; E = $2+10 = 12$.
- Settle B (3): D = $\min(10, 3+5) = 8$; (C already settled). - Settle D (8): E = $\min(12, 8+2) = 10$; F = $8+6 = 14$.
- Settle E (10): F = $\min(14, 10+3) = 13$. - Settle F (13): done.

Shortest path A $\rightarrow$ F: A $\rightarrow$ C $\rightarrow$ B $\rightarrow$ D $\rightarrow$ E $\rightarrow$ F

Total distance: 13.

(Check: A $\rightarrow$ C 2, C $\rightarrow$ B 1, B $\rightarrow$ D 5, D $\rightarrow$ E 2, E $\rightarrow$ F 3 = $2+1+5+2+3 = 13$. Compare alternative A $\rightarrow$ C $\rightarrow$ B $\rightarrow$ D $\rightarrow$ F = $2+1+5+6 = 14$, longer. So 13 is optimal.)

Challenge

(a) A heuristic is **admissible** if it **never overestimates** the true shortest-distance from a node to the goal F. To check, compare each $h(n)$ to the actual shortest distance to F (computed by Dijkstra from each node): - True shortest distances to F — F:0, E:3, D:5 (D $\rightarrow$ E $\rightarrow$ F = $2+3$), B:10 (B $\rightarrow$ D $\rightarrow$ E $\rightarrow$ F = $5+2+3$), C:11 (C $\rightarrow$ B $\rightarrow$ D $\rightarrow$ E $\rightarrow$ F = $1+5+2+3$), A:13. - Heuristic given: A=9, B=7, C=8, D=4, E=3, F=0. Each is $\leq$ the true distance ($9 \leq 13$, $7 \leq 10$, $8 \leq 11$, $4 \leq 5$, $3 \leq 3$, $0 \leq 0$), so the heuristic is **admissible** and A* would find the optimal path.

(b) A explores fewer nodes than Dijkstra when a good (informative) heuristic directs the search toward the goal instead of expanding outward in all directions — useful when you only need the path to one goal node. A behaves **exactly like Dijkstra** when the heuristic is **$h(n) = 0$** for every node (no information), because then $f(n) = g(n)$.